

homework-2

Question 1

```
install.packages("aRtsy")
```

Installing the package using the `install.packages()` will permanently download the package to your computer, while the `library()` will load it into your R Studio session's environment allowing you to use the functions and information in the package.

Packages

Question 2

```
library("aRtsy")
```

Essentially, the Collatz Conjecture is a piece-wise function that outputs the value one no matter what positive number you initially put into the equation.

The Collatz Conjecture

$$f(n) = \begin{cases} n \div 2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$

Question 3

a.

The function plots the x and y-values generated by the piecewise function onto a canvas in RStudio. The resulting graph shows drastic increases and decreases in the y-values as the function goes through each step of the Collatz Conjecture.

b.

- I get a different output each time I enter a different number because the sequence of equations required to reach 1 is different for every positive number.
- It would be difficult to replicate the collatz conjecture code in a .qmd file because every output that is not 1 would need to be passed through its corresponding equation repeatedly until the sequence eventually reaches 1.
- By setting the seed to a certain integer, you are able to get the same output of numbers using the sample function depending on how many numbers you want to output and through which numbers you want to index. Your output of numbers will also vary depending on which functions you use. i.e.(sample, rnorm)

sample

```
set.seed(32)
sample(1:100, 3)
```

```
[1] 70 11 20
```

```
set.seed(32)
sample(1:100, 3)
```

```
[1] 70 11 20
```

```
set.seed(17)
sample(1:100, 3)
```

```
[1] 50 97 94
```

rnorm

```
set.seed(32)  
rnorm(3)
```

```
[1] 0.01464054 0.87328871 -1.02794620
```

```
set.seed(32)  
rnorm(3)
```

```
[1] 0.01464054 0.87328871 -1.02794620
```

```
set.seed(16)  
rnorm(3)
```

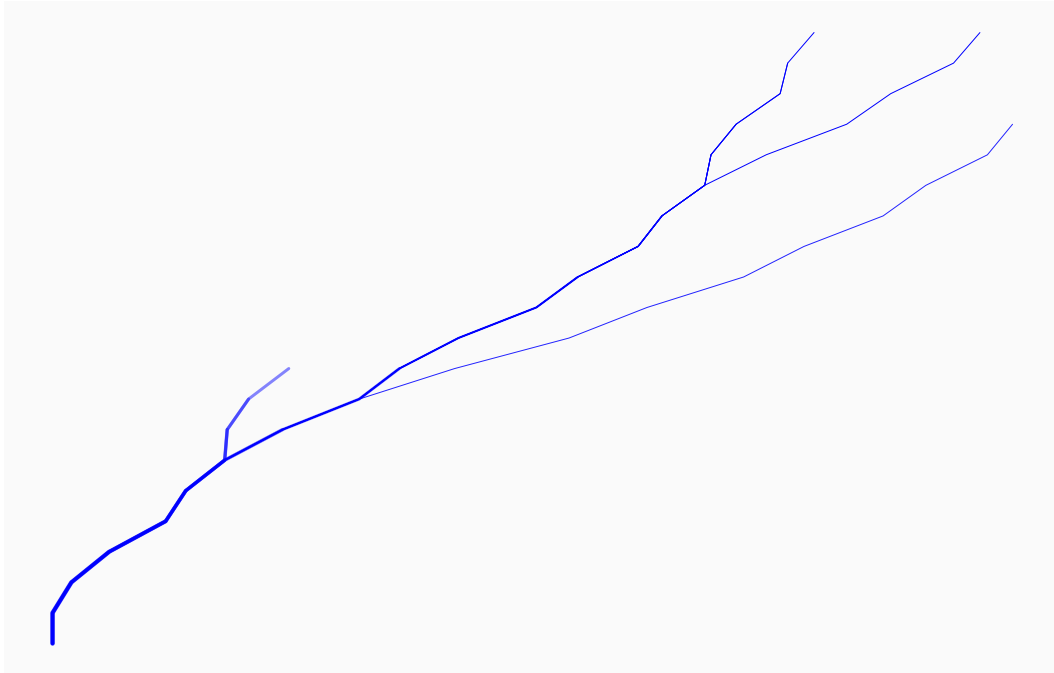
```
[1] 0.4764134 -0.1253800 1.0962162
```

Question 4

```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```



```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



The code in the previous chunk is substituting the value `n` only for 10, while the code in question four is substituting `n` by indexing all number 1 through 20. The outcome, the code in question four is printing the `n`-values 1:20 onto the canvas while the code from the previous chunk is only printing the outcome of the `n`-value 10. While using the `set.seed` function to set the seed to 1 does change the starting value to one, the index already starts from the number 1, so it does not make any sense to set the seed to start at 1.